The Impact of Industrial Opt-Out from Utility Sponsored Energy Efficiency Programs

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Abstract

Industry accounts for one-third of energy consumption in the US. Studies suggest that energy efficiency opportunities represent a potential energy resource for regulated utilities and have resulted in rate of return regulated demand-side management (DSM) and energy efficiency (EE) programs. However, many large customers are allowed to self-direct or opt-out. In the Carolinas (NC and SC), over half of industrial and large commercial customers have selected to opt out. Although these customers claim they invest in EE improvements when it is economic and cost-effective to do so, there is no mechanism to validate whether they actually achieved energy savings. This project examines the industrial energy efficiency between the program participants and non-participants in the Carolinas by utilizing the non-public Census of Manufacturing data and the public list of firms that have chosen to opt out. We compare the relative energy efficiency between the stay-in and opt-out plants. The t-test results suggest opt-out plants are less efficient. However, the opt-out decisions are not random; large plants or plants belonging to large firms are more likely to opt out, possibly because they have more information and resources. We conduct a propensity score matching method to account for factors that could affect the opt-out decisions. We find that the opt-out plants perform at least as well or slightly better than the stay-in plants. The relative performance of the opt-out firms suggest that they may not need utility program resources to obtain similar levels of efficiency from the stay-in group.

Keyword*:* Demand-side management, energy efficiency, energy policy

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1 Introduction

Industry accounts for about one-third of energy consumption in the US (US EIA, 2021). Studies suggest that potential energy efficiency[1] opportunities represent an energy resource for regulated utilities and have resulted in rate of return regulated demand-side management (DSM) and energy efficiency (EE) programs (Chittum 2011, Scull-MEEA 2017). However, many businesses complain about the value of these programs and associated higher rates, so many public utility commissions (PUC) in the U.S. have changed the participation provisions for business customers regarding these programs (Chittum and Elliott, 2009). In the "Carolinas" (North and South Carolina), this comes in the form of a formal "opt-out" process in the regulated electricity rates. Industrial and large commercial customers, with annual energy usage of more than a million kWh, may opt out. Customers notify utilities and attest that they have or will implement alternative DSM and EE measures. Opt-out customers do not have access to program resources and are not required to pay the cost recovery rider (NCUC Docket No E-7, PSCSC Docket No. 2020-83-E). Over half of the eligible customers have selected to opt out (Weiss et al., 2017). Although these customers attest to EE improvements when it is economic and cost-effective to do so, there is no requirement to validate EE measures. This project examines the relative energy efficiency between the program participants (stay-ins) and non-participants (opt-outs) in the Carolinas utilizing two novel datasets. The first is non-public data from Census of Manufactures (CMF) and the second is data compiled from the public regulatory docket. We first conduct a national panel analysis using non-public data to measure industry sector specific plant-level energy efficiency and then com-

[1]Throughout this paper, "energy efficiency" will refer to both electricity and natural gas efficiency in broad discussion. Although energy efficiency programs apply to both electricity and natural gas sectors, they are more common in regulated electricity sector, compared to natural gas. In this paper, we focus on electricity measures in our analyses.

pare those estimates of energy efficiency between stay-in and opt-out plants based on matched public/non-public data from the Carolinas.

The paper contributes to the literature in three fundamental ways. The first is to create a dataset of companies that choose to opt-out of the programs, but this does not provide any information on their performance, or the performance of those that stay-in. Even if data were available from the regulated utilities on energy use, the best one could do would be to measure the amount of energy associated with the two groups. Without a measure of productive activity in the plant there is no way to measure energy intensity. We address this problem by using non-public data from the U.S. Census Bureau. The second fundamental contribution is to take plant level energy intensity and create a measure of energy efficiency by comparing plants' energy intensity to a national peer group. Again, data just from the Carolinas would be too sparse for this purpose. Using the residuals from a regression analysis we create an industry specific measure of relative energy intensity. Finally the study considers the notion that the decision to opt-out is not random. Average treatment effects are estimated to capture this.

Our project also employs two novel data sources linked at the firm level. The first is the non-public, plant-level from the quinquennial CMF, which is used to measure plant-level energy efficiency, controlling for observable industry specific heterogeneity. These data are only accessible with special permission from the Census Bureau. The CMF data provide information on plant-level energy use, prices, and production needed to measure energy efficiency. The second dataset is compiled from the public regulatory docket on firms that opt-out of the EE and DSM rates. Duke Energy Progress and Duke Energy Carolinas[2] have over 1,000 opt-out industrial firms listed

[2]In NC and SC, Duke Energy Progress, Duke Energy Carolinas, and Dominion Energy are the regulated entities that operate the energy efficiency and demand-side management programs and associated energy rates. Duke Energy Progress and Duke Energy Carolinas have the opt-out list publicly available and we utilize this public list to compare energy efficiency in this paper.

in their public filings. We use firm names to merge the non-public CMF with the public opt-out data.

In the first stage of national analyses, we compute the energy inefficiency measures for each plant in the entire country. For each 6-digit NAICS code, we regress electricity use on the electricity price, plant-level activity (production, labor, and materials) and additional controls (state average natural gas price and weather information). Average plant-level electricity prices are computed from total reported consumption and expenditures. Shipment values are our production measure; labor is total production worker hours; cost of materials captures possible energy embodied in other inputs. We also include state-year average natural gas (US Energy Information Administration) and weather information (cooling degree-days linked via plant zip code). The years 2002, 2007, 2012, and 2017 are included in our national panel analyses. We estimate the models with OLS and state / year fixed effects. One concern with OLS is price endogeneity. Large energy users may have means to get the lower electricity prices and OLS specifications may result in biased electricity price elasticity. We use an IV estimate with the state-year average electricity price as an instrument for plant-level electricity price. Energy inefficiency measures are computed from the regression residuals of both OLS and IV. Since all variables are in logs, if the residual is positive, the plant uses relatively more energy than predicted and we define such a plant as relatively inefficient.

After computing the inefficiency measures, we compare the relative energy inefficiency between the stay-in and opt-out plants within these two states. We first conduct a simple t-test and compare the means of inefficiency for stay-in and opt-out plants. However, opt-out decisions may not be random: for example, large plants or plants belonging to large firms may be more likely to opt out possibly because they have more information and resources. We conduct a propensity score matching

(PSM) method to account for factors that could affect the opt-out decisions. In the PSM choice model, we consider plant-level attributes, firm size (total national shipment values and dummies indicating whether the firm has multiple plants in the US and in the state that the plant is located). We compute the average treatment effect (ATE) and calculate how opting out may affect the plant inefficiency.

Using both the OLS and IV efficiency estimates, the t-test shows opt-out plants have significantly higher mean energy inefficiency measures and therefore are more inefficient. However, the PSM choice model shows larger plants and plants belonging to large firms are more likely to opt out. Plants belonging to firms with multiple plants in the state are also more likely to opt out, indicative of a local network effect. Compared to the stay-in plants, the ATE results find that the opt-out plants perform at least as well (using IV based efficiency estimates) or slightly better (using OLS based efficiency estimates). We are unable to conclude from this analysis whether utility programs are effective, since we do not observe which members of the stay-in group elects to participate in the programs. It may be that firms that opt-out are more aware of their utility costs and more actively manage their energy, even given that the stay-in plants have access to utility provided resources. PSM model shows that the decision to opt out is not random and affects the comparison between the two groups. While this study cannot determine the effectiveness of the DSM and EE programs, the relative performance of the opt-out firms suggest that they do not appear to need utility program resources to obtain similar levels of efficiency as the stay-in group.

The remainder of the paper is organized as follows. Section 2 provides information about energy efficiency programs and opt-out provisions generally, then specific details about the Carolinas. Section 3 describes the two data sets. Section 4 and 5 provide the estimation details and empirical results. Section 6 includes concluding observations.

2 Demand-Side Management and Energy Efficiency Programs

2.1 Program Background

Despite increased energy efficiency of electric devices, energy demand and consumption in- creases every year. The US will face a number of challenges, including aging plants, constraints on existing transmission and distribution systems, environmental regulations, and grid capacity concerns (Palensky and Dietrich 2011, Chittum 2011). Industrial sector accounts for almost one third of all energy consumption in the US (US EIA, 2021) and this represents a large opportunity for energy savings. Two common responses to increasing energy needs are reducing energy demand and acquiring new energy supply (Chittum, 2011). Compared to developing new energy resources, reducing energy demand through demand-side management (DSM) or energy efficiency (EE) programs is only 1/10 to 1/3 of the cost of new renewable and non-renewable energy generation resources (Friedrich et al. 2009, Relf et al. 2020) and such low cost makes energy efficiency as the cheapest energy resource.

Because of the low cost and speed, for example, EE measures could result in immediate and permanent energy savings (Palensky and Dietrich, 2011), many states and utilities viewed EE as a priority when planning for future energy demand (Chittum, 2011) and have been implementing DSM and EE programs since 1970s (Auffhammer et al., 2008). Spending on EE programs has increased every year and total budgets in 2010 has increased to \$6.5billion (Molina et al., 2010).[3] The programs generally employ a cost-recovery mechanism (CRM). Customers need to pay additional fee, for example DSM/EE rider, and the aggregated funds are pooled together to recover all reasonable and prudent costs incurred in the programs (Chittum, 2011). These

[3]According to (Chittum, 2011), over 41 states have implemented some CRM programs

programs aim to improve the energy system, reduce energy demand, and improve energy efficiency through dissemination programs, subsidies, and free installation of more efficient technologies and equipment (Auffhammer et al. 2008, Palensky and Dietrich 2011). Other offers include energy audits, technical assistance, financial incentives, and rebates for investments in EE equipment or adoption of EE behaviors (Chittum, 2011). Utilities also innovate over time to meet changing system needs. For example, various utilities piloted new programs in 2018, including smart thermostats, online marketplaces for EE products, and demand response and storage systems. In 2015, very few utilities implemented geo-targeted efficiency, and the number had only marginally increased by 2018 (Relf et al., 2020).

Because each state has its own standards, regulations and programs[4] and state policies may vary over time. For example, according to 2020 State Energy Efficiency Scorecard (Berg et al., 2021), California has become the leader in saving energy. It has implemented net-zero energy building codes, low-emissions and zero-emission vehicle regulations. Iowa fell the farthest in the EE scorecard ranking because its 2018 legislation capped demand-side investment and allowed customers to opt out programs. It is also the state that fell most in 2017 scorecard (Berg et al., 2018) due to the temporary suspension of funding for efficiency loan and grant programs.[5] The performance in energy efficiency programs may be related to state policies and regulatory support: the ACEEE's Utility Energy Efficiency Scorecard report (Relf et al., 2020) shows all of the top 10 utilities are located in states ranking top 13 in ACEEE's State Energy Efficiency Scorecard.

[4] According to Berg et al. 2021, 27 states have adopted a utility-sector EE resource standard.
[5] According to 2020 scorecard, NC ranks 27 and SC ranks 40.

2.2 Alternative Options

Large customers generally support opt-out or self-direct options when such public benefits fund programs are in place as these customers argue that they do not receive equal benefits of the costs they contribute to the cost-recovery management (CRM) funding. Because the CRM fees are typically based on a percentage of the customer's monthly bill, energy-intensive industrial firms usually contribute more to the funding pools. They also assert that these programs may not be responsive to their needs and they have to pay for services that they do not expect to use. Because of the economic pressure to reduce energy cost, large customers suggest they should be able to spend the CRM funding to invest their own cost-effective energy efficiency projects and build their own EE measures as they know best the internal realities of their firms and can make best decisions on their capital investments (Chittum and Elliott 2009, Chittum 2011).

Many utilities now offer the opt-out and self-direct options.[6] Such options are growing in popularity. These programs usually affect industrial customers, but many have included other large customers as well (Baatz et al., 2017), using some specified annual kWh consumption minimum threshold (Chittum, 2011). Customers who have chosen to opt out generally are exempt from paying riders and as a result, they no longer have access to the program resources (Baatz et al., 2017). Self-direct programs allow customers to self-direct some or all of their CRM funding to their own EE measures (Chittum, 2011). One problem with the opt-out and self-direct policies is there is often no formal mechanisms validating they have installed their own EE measures. Although industrial customers are likely aware of EE projects in their facilities, many small or medium-sized projects usually cannot compete with other capital demands

[6] According to (Chittum, 2011), 23 out of 41 states have some opt-out or self-direct options in their programs

and this could result in low EE participation (Chittum and Elliott 2009, Weiss et al. 2017). Studies found many industrial firms have not captured many cost- effective energy saving opportunities. One consequence from the low EE participation is that the opted-out customers could be free riders who receive the EE savings benefits at no cost, while this imposes an increasing rider cost for those participating customers (Weiss et al., 2017). The opt-out policy for large customers could increase demand and electricity costs for customers. Large customers typically comprise a majority of energy demand and energy savings opportunity, and such savings can be the lowest cost to acquire. The loss of energy savings from large customers could result in decrease in the reductions in infrastructure costs and increasing costs for all customers. The higher energy demand increases wholesale energy prices and raises the costs for all customers. One study focusing on Ohio found the policies result in higher demand, increased air pollution, and lost bill savings opportunities from large customers, which lead to higher utility system investments, health costs, and lost economic growth (Baatz et al., 2017).

2.3 Carolinas' Options

In 2007, the NC legislature established the Renewable Energy and Energy Efficiency Portfolio Standard (REPS) in Senate Bill 3. REPS requires each utility to increase the percentage of energy mix from renewable sources over time and allows a portion to be met with energy efficiency (Weiss, 2019). For SC, the South Carolina General Assembly passed the South Carolina Energy Efficiency Act and created the State Energy Office in 1992. This act created the state energy plan and since then, SC has included EE in energy and public building conservation plans (Weiss, 2021). Utilities (mainly investor-owned utilities, including Duke Energy Carolinas (DEC), Duke Energy Progress (DEP), and Dominion Energy) have helped to encourage EE investment by

providing DSM and EE programs, such as EE rebates and incentives (Weiss, 2019). In the Carolinas, firms served by Duke Energy automatically stay in the programs and pay cost recovery riders.[7]

Utilities help customers navigate the rebate process, prioritize energy savings investment, and assign large account managers to firms with $250,000 annual revenue even if some plants use fewer than 1M meters. Their Virtual Energy Assessment team offers no-cost analysis customized to each project and to help customers make decisions about energy efficiency improvements. Another program, Smart $aver Customer Incentive program, helps customers improve energy efficiency and offers incentives to allow the customers to offset the cost of upgrades. Some common upgrades include high-efficiency lighting, air compressors and controls modernization, induction heaters and control sequences, and heavy metal forming equipment replacement. The program creates special EE improvement projects for each customer, which are tailored to the customers' unique organization and needs.

Despite the help offered by the programs, large industrial customers lobby to opt out and suggest they should build their own programs. In the Carolinas, any industrial, regardless of energy use, and large customers (commercial/county/school), which are defined as the ones have annual energy usage more than 1,000,000 kilowatt hours, are allowed to opt out DSM and EE programs[8] (NCUC Docket No E-7, PSCSC Docket

[7]For example, in NC DEC 2017, EE rider is 0.0088¢ per kWh and 0.0014¢ per kWh for DSM. In SC, the EE rider is 0.0126¢ and 0.0009¢ per kWh for DSM (DEP and DEC Documents)

[8]Residential customers automatically opt in and have no options to opt out.

No. 2020-83-E). To opt out, the customer needs to submit written documents to the utilities and demonstrate between November 1st and December 31st that it has or will implement alternative DSM or EE measures in accordance with quantifiable goals and they will start to opt out in the coming January 1st (SEEA 2019, NCUC Docket No.E-7). The customer is defined as a metered account billed under a single application of a company rate tariff. Hence, a firm with several meters could have some opted-out while the others stay in. Once the customers decide to opt out, they opt out of the utilities' entire portfolio of programs. In return, they do not have access to resources and are not required to pay the cost recovery rider (NCUC Docket No E-7, PSCSC Docket No. 2020-83-E). Over half of industrial and large commercial customers have chosen to do so (Weiss et al., 2017). In 2018, opted-out customers account for half of eligible sales in Duke Energy's territories (Weiss, 2019). Customers are allowed to re-participate in a new utility-provided DSM or EE programs. DEC requires customers to opt in for one year and DEP requires them to opt in for three years (SEEA, 2019).

3 Data

3.1 National Analyses

We used plant-level information from the non-public Census of Manufactures (CMF) in the national analyses when compute the energy efficiency measures. The data are protected under Title 13 and 26 of the US code and are used under permission of Census Bureau. The CMF data are part of the quinquennial Economic Census and cover virtually all plants in the industry. We use plant-level electricity use and prices, which are calculated from electricity consumption and costs. We include plant-level shipment values as production measure. Labor is measured by total production hours and we also include cost of materials as additional control. We include years 2002,

2007, 2012, and 2017 in our national analyses. Additional controls are state-year average natural gas price and weather information (cooling degree days), both provided by the US Energy Information Administration (EIA).

3.2 Duke Energy Opt-Out List

DEP and DEC have the opt-out lists publicly available. The lists include the firms and the corresponding accounts that opted out DSM and EE programs between the years 2016 and 2019. These lists include all kinds of customers, including industrial, commercial, school, and other large facilities. However, in order to merge with the CMF data, we only consider the industrial firms in our analyses. We use the firm names to merge the CMF and opt-out data. A firm is defined as opted-out if it has any accounts opted out in either DSM or EE programs in DEP or DEC.

Because of the opt-out policies, firms may opt in and out year by year. We analyzed the opt out list and found that very few customers change their plans over time.

4 National Analyses

In the first stage, we national analyses for each 6-digit NAICS code:

$$log(\mathrm{E}_{i,t}) = \beta_0 + \beta_1 log(\mathrm{P}_{i,t}) + \beta_2 log(\mathrm{Y}_{i,t}) + \beta_3 log(\mathrm{L}_{i,t}) + \beta_4 log(\mathrm{M}_{i,t}) + \beta_5 log(\mathrm{PrF}_{i,t})$$

$$+ \beta_6 \mathrm{CDD}_{i,t} + \beta_7 \mathrm{State}_i + \beta_8 \mathrm{Year}_i + \varepsilon_{i,t}$$

where the dependent variable is (logged) electricity use, and the independent variables are (logged) output, worker production hours, cost of materials, state-average natural gas price, and cooling degree days.

We conduct two kinds of analyses. In the simple OLS estimates, we include the state fixed effects as each state has its own programs and alternative options. We also include year fixed effects to control for time trends. Price endogeneity is a concern in the analyses. Because large energy users want the lowest possible energy prices, using simple OLS may bias the price elasticity upwards in absolute terms, we consider state-year average electricity price as the instrument.

After the national regressions, we compute the residuals as the inefficiency measures. If the plant has a positive residual, we define this as inefficient since the plant uses more energy than predicted.

5 Inefficiency Between Stay-in and Opt-out Plants

After calculating the inefficiency measures, we first conduct a t-test and compare the means of inefficiency between stay-in and opt-out firms.[9] The difference is defined as the mean of stay-in plants minus the one of opt-out plants.

Table 1: T-test: Stay-in versus Opt-out

	IV	OLS
diff	-0.1124***	-0.09127**
	(-3.121)	(-2.330)
N	9700	9700

Standard errors in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Both IV and OLS approaches show that opt-out plants have more negative inefficiency means and hence are more inefficient.

However, the opt-out decisions may not be random. For example, large plants or plants belonging to large firms may be more likely to opt out as they have more information and resources. We therefore conduct a propensity score matching analysis.

[9]Because Duke Energy's public opt-out list includes firms between years 2016 and 2019, we only include year 2017 information and only plants in NC and SC.

In the choice model, we run a probit regression:

$$\begin{aligned}\text{Opt-Out}_i =& \beta_0 + \beta_1 log(\text{E}_i) + \beta_2 log(\text{Y}_i) + \beta_3 log(\text{Y}_{nation,i}) \\ &+ \beta_4 \text{MP}_{nation,i} + \beta_5 \text{MP}_{state,i} + \beta_6 \text{State}_i + \varepsilon_i\end{aligned}$$

where the dependent variable is a dummy that equals one if the plant opts out. The independent variables include logged electricity use, logged output, logged national-summed output,[10] two dummies indicating whether the firm has multiple plants in the US and the state.[11] Because the Carolinas include two states, we include a state dummy equal to one if the plant is located in NC.

Table 2: PSM - Choice Model

	Probit
log(E)	0.1730***
	(0.01894)
log(Y)	-0.05910**
	(0.02669)
$log(\text{Y}_{\text{nation}})$	0.1578***
	(0.01568)
$\text{MP}_{\text{nation}}$	0.06708
	(0.08040)
MP_{state}	0.5956***
	(0.06488)
State	0.9224***
	(0.05813)
N	9700
Pseudo R^2	0.3701

Standard errors in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

The results in the choice model suggest that larger plants (such as those with

[10]This variable sums the output of all plants for each firm in year 2017
[11]Variables equal one if the firm has more than one plant in NC/SC or in US in year 2017.

higher output or energy intensity), plants belonging to large firms, or plants belonging to firms with multiple plants in the state are more likely to opt out. We then calculate the average treatment effect in the PSM model and test how opt out affects inefficiency.

Table 3: PSM - ATE Stay-in versus Opt-out

	IV	OLS
ATE	-0.2059	-0.2422*
	(-1.483)	(-1.740)
N	9700	9700

Standard errors in parentheses
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

The results are opposite from the t-test: in the IV efficiency estimates, there is no significant difference between stay-in and opt-out plants and in the OLS approach, the opt-out plants perform slightly better (less inefficient) but the p-value is not very large.

6 Conclusion and Policy Implications

EE has been demonstrated to be a cost effective resource. For this reason most regulated utilities have some form of EE programs. However, the complexity of industrial energy use may lead to a mis-match of the public and private provision of EE. While EE has aspects of a public good, private companies may act in their own private interests to manage energy use. Since there is ample evidence that a wide range of performance can exist for manufacturing energy use, those that are already energy efficient by their own actions might end up paying for EE programs without any direct benefits. In other words, these firms might provide the public benefit from their own actions and should not be expected to pay the public benefit rider. The problem with this argument is that most opt-out provisions have no requirement to

validate the firms' EE efforts. This study sheds light on this by using novel data on public opt-out decisions and non-public data on energy efficiency.

In a simple comparison we find that firms that opt-out are more inefficient on average, but this is misleading, since the decision to opt-out is not random. When this is accounted for in the estimation procedure, firms that opt-out have a similar level of efficiency as those that stay in . This suggests that, when compared to an appropriate peer group, firms that opt-out appear to be implementing EE actions similar to those in the stay-in. In other words, the opt-out group appears to be providing a similar level of public benefit via their own self-interested actions.

However, the data we observe for the "stay-in group" does not include information about the impact of EE programs on those firms that actively choose to participate in the program offerings. Our data do not allow us to observe firms that take advantage of the EE program offerings versus those that choose not to or are simply unaware of the program offerings. This is likely to increase the variance of the efficiency estimates for the stay-in group and complicate the underlying inference between stay-in and opt-out.

While this study sheds light on those firms that actively choose to not participate in funded EE programs via opt-out, it cannot assess the effectiveness of those programs on those that actively choose to participate versus those that are unaware of the program or the ability to opt-out. This would require data from the utilities on program participants. Such data could either come from the PUC requiring this data to be made public, or an agreement with the utility companies on sharing that data for research purposes. This would be a fruitful extension of this work for future research.

CRediT authorship contribution statement

Gale Boyd: Conceptualization, Data acquisition, Methodology, Formal analysis, Writing - review & editing, Funding acquisition, Supervision. **Matthew Doolin**: Data curation. **Yu Ma**: Data curation, Methodology, Formal analysis, Writing - original draft, Writing - review & editing. **Jennifer Weiss**: Conceptualization, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The census data that has been used is confidential. Upon request, the authors agree to assist any formal requests to the Federal Statistical Research Data Center (FSRDC) program for access to the confidential portions of the datasets. The public lists of firms that have chosen to opt out are publicly available. The lists include both manufacturing and non-manufacturing customers. In order to merge with Census of Manufactures data, we only include manufacturing firms in this project.

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References

Auffhammer, Maximilian, Blumstein, Carl, and Fowlie, Meredith. "Demand-side management and energy efficiency revisited." *The Energy Journal* 29.3 (2008).

Baatz, Brendon J, Relf, Grace, and Kelly, Meegan. "Consequences of large customer opt-out: An Ohio example." *The Electricity Journal* 30.9 (2017): 29–36.

Berg, Weston, Nowak, Seth, Kelly, Meegan, Vaidyanathan, Shruti, Shoemaker, Mary, Chittum, Anna, DiMascio, Marianne, and Kallakuri, Chetana. "The 2017 State Energy Efficiency Scorecard, American Council for an Energy-Efficient Economy." Tech. rep., Report, 2018.

Berg, Weston, Vaidyanathan, Shruti, Junga, E, Cooper, Emma, Perry, Chris, Relf, G, Whitlock, A, DiMascio, M, Waters, C, and Cortez, N. "The 2020 State Energy Efficiency Scorecard, American Council for an Energy-Efficient Economy." *American Council for an Energy Efficient Economy. Washington DC* .

Chittum, Anna. "Follow the Leaders: Improving Large Customer Self-Direct Programs." American Council for an Energy-Efficient Economy, 2011.

Chittum, Anna and Elliott, Neal. "Implementing Industrial Self-Direct Options: Who Is Making It Work?" *Proceedings of the 2009 ACEEE Summer Study on Energy Efficiency in Industry* .

Friedrich, Katherine, Eldridge, Maggie, York, Dan, Witte, Patti, and Kushler, Marty. "Saving energy cost-effectively: a national review of the cost of energy saved through utility-sector energy efficiency programs." *American Council for an Energy-Efficient Economy. September* .

Molina, Maggie, Neubauer, Max, Sciortino, Michael, Nowak, Seth, Vaidyanathan, Shruti, Kaufman, Nate, Chittum, Anna, Sheppard, Colin, Harper, Margaret, Jacobson, Arne, et al. "The 2010 state energy efficiency scorecard." 2010.

Palensky, Peter and Dietrich, Dietmar. "Demand side management: Demand response, intelligent energy systems, and smart loads." *IEEE transactions on industrial informatics* 7.3 (2011): 381–388.

Relf, Grace, Cooper, Emma, Gold, Rachel, Goyal, Akanksha, and Waters, Corri. "2020 Utility Energy Efficiency Scorecard." *Washington, DC: American Council for an Energy Efficiency Economy* .

Scull-MEEA, Leah. "Making the Case for Inclusive Industrial Energy Efficiency Policy-By Leah Scull." .

SEEA. "State Guide to Utility Energy Efficiency Planning: North Carolina." *SOUTHEAST ENERGY EFFICIENCY ALLIANCE* .

Weiss, Jennifer. "North Carolina Energy Efficiency Roadmap: Executive Summary. NI R 19-03-03." *Nicholas Institute for Environmental Policy Solutions. Durham, NC: Duke University* .

———. "South Carolina Energy Efficiency Roadmap." .

Weiss, Jennifer, Stephens, Perry, and Stowe, Michael. "Realigning Utility Incentives for Industrial Customers: Using Strategic Energy Management to Increase Customer Participation in Energy Efficiency Programs." *American Council for an Energy Efficient Economy. Washington DC* .